

Question 01

- a)
- i. Create a database named hotel_db
- Query:
- ```
CREATE DATABASE hotel_db;
USE hotel_db;
```
- ii. Create the tables with appropriate primary key and foreign key with the suitable data types.

Query:

```
CREATE TABLE Hotel (
 HotelNo VARCHAR(5) PRIMARY KEY,
 Name VARCHAR(25),
 City VARCHAR(15)
);
```

```
CREATE TABLE Guest (
 GuestNo INT PRIMARY KEY,
 GuestName VARCHAR(25),
 GuestAddress VARCHAR(50)
);
```

```
CREATE TABLE Room (
 RoomNo INT,
 HotelNo VARCHAR(5),
```

```
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Type VARCHAR(20),
Price DECIMAL(5,2),
PRIMARY KEY (HotelNo, RoomNo),
FOREIGN KEY (HotelNo) REFERENCES Hotel(HotelNo)
);
```

```
CREATE TABLE Booking (
HotelNo VARCHAR(5),
GuestNo INT,
DateFrom DATE,
DateTo DATE,
RoomNo INT,
PRIMARY KEY (HotelNo, GuestNo, DateFrom),
FOREIGN KEY (HotelNo) REFERENCES Hotel(HotelNo),
FOREIGN KEY (GuestNo) REFERENCES Guest(GuestNo),
FOREIGN KEY (RoomNo, HotelNo) REFERENCES Room(RoomNo, HotelNo)
);
```

```

MariaDB [hotel_db]> CREATE TABLE Hotel (
 -> HotelNo VARCHAR(5) PRIMARY KEY,
 -> Name VARCHAR(25),
 -> City VARCHAR(15)
 ->);
Query OK, 0 rows affected (0.021 sec)

MariaDB [hotel_db]> CREATE TABLE Guest (
 -> GuestNo INT PRIMARY KEY,
 -> GuestName VARCHAR(25),
 -> GuestAddress VARCHAR(50)
 ->);
Query OK, 0 rows affected (0.015 sec)

MariaDB [hotel_db]> CREATE TABLE Room (
 -> RoomNo INT,
 -> HotelNo VARCHAR(5),
 -> Type VARCHAR(20),
 -> Price DECIMAL(5,2),
 -> PRIMARY KEY (HotelNo, RoomNo),
 -> FOREIGN KEY (HotelNo) REFERENCES Hotel(HotelNo)
 ->);
Query OK, 0 rows affected (0.015 sec)

MariaDB [hotel_db]> CREATE TABLE Booking (
 -> HotelNo VARCHAR(5),
 -> GuestNo INT,
 -> DateFrom DATE,
 -> DateTo DATE,
 -> RoomNo INT,
 -> PRIMARY KEY (HotelNo, GuestNo, DateFrom),
 -> FOREIGN KEY (HotelNo) REFERENCES Hotel(HotelNo),
 -> FOREIGN KEY (GuestNo) REFERENCES Guest(GuestNo),
 -> FOREIGN KEY (RoomNo, HotelNo) REFERENCES Room(RoomNo, HotelNo)
 ->);

```

iii. Insert some sample data into the table's hotel, rooms, booking and guest as given below:

INSERT INTO Hotel (HotelNo, Name, City) VALUES

('ch01', 'Omni Shoreham', 'London'),

('ch02', 'Phoenix Park', 'London'),

('dc01', 'Latham', 'Berlin'),

('fb01', 'Grosvenor', 'London'),

('fb02', 'Watergate', 'Paris');

INSERT INTO Guest (GuestNo, GuestName, GuestAddress) VALUES

(10001, 'John Kay', '56 High St, London'),

(10002, 'Mike Ritchie', '18 Tain St, London'),

(10003, 'Mary Tregear', '5 Tarbot Rd, Aberdeen'),

(10004, 'Joe Keogh', '2 Fergus Dr, Aberdeen'),

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(10005, 'Carol Farre', '16 Achray St, Glasgow'),

(10006, 'Tine Murphy', '63 Well St, Glasgow'),

(10007, 'Tony Shaw', '12 Park Pl, Glasgow');

INSERT INTO Room (RoomNo, HotelNo, Type, Price) VALUES

(501, 'fb01', 'single', 19.00),

(601, 'fb01', 'double', 29.00),

(701, 'ch02', 'single', 10.00),

(701, 'fb01', 'family', 39.00),

(801, 'ch02', 'double', 15.00), (901,

'dc01', 'single', 18.00),

(1001, 'ch01', 'single', 29.99),

(1001, 'dc01', 'double', 30.00),

(1001, 'fb02', 'single', 58.00),

(1101, 'ch01', 'family', 59.99),

(1101, 'dc01', 'family', 35.00),

(1101, 'fb02', 'double', 86.00);

INSERT INTO Booking (HotelNo, GuestNo, DateFrom, DateTo, RoomNo) VALUES

('ch01', 10006, '2004-04-21', NULL, 1101),

('ch02', 10002, '2004-04-25', '2004-05-06', 801),

('dc01', 10003, '2004-05-20', NULL, 1001),

('dc01', 10007, '2004-05-13', '2004-05-15', 1001),

('fb01', 10001, '2004-04-01', '2004-04-08', 501),

('fb01', 10001, '2004-05-01', NULL, 701),

('fb01', 10002, '2004-05-04', '2004-05-29', 601),

('fb01', 10004, '2004-04-15', '2004-05-15', 601),

('fb01', 10005, '2004-05-02', '2004-05-07', 501),

('fb02', 10003, '2004-04-05', '2010-04-04', 1001),

('fb02', 10005, '2004-05-12', '2030-05-04', 1101);

```

MariaDB [hotel_db]> INSERT INTO Hotel (HotelNo, HotelName, City) VALUES('ch01', 'Omni Shoreham', 'London'),('ch02', 'Phoenix Park', 'London'),('dc01', 'Latham', 'Berlin'),('fb01', 'Grosvenor', 'London'),('fb02', 'Watergate', 'Paris');
Query OK, 5 rows affected (0.005 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [hotel_db]> INSERT INTO Guest (GuestNo, GuestName, GuestAddress) VALUES(10001, 'John Kay', '56 High St, London'),(10002, 'Mike Ritchie', '18 Tain St, London'),(10003, 'Mary Tregear', '5 Tarbot Rd, Aberdeen'),(10004, 'Joe Keogh', '2 Fergus Dr, Aberdeen'),(10005, 'Carol Farre', '16 Achray St, Glasgow'),(10006, 'Tine Murphy', '63 Well St, Glasgow'),(10007, 'Tony Shaw', '12 Park Pl, Glasgow');
Query OK, 7 rows affected (0.005 sec)
Records: 7 Duplicates: 0 Warnings: 0

MariaDB [hotel_db]> INSERT INTO Room (RoomNo, HotelNo, Type, Price) VALUES(501, 'fb01', 'single', 19.00),(601, 'fb01', 'double', 29.00),(701, 'ch02', 'single', 10.00),(701, 'fb01', 'family', 39.00),(801, 'ch02', 'double', 15.00),(901, 'dc01', 'single', 18.00),(1001, 'ch01', 'single', 29.99),(1001, 'dc01', 'double', 30.00),(1001, 'fb02', 'single', 58.00),(1101, 'ch01', 'family', 59.99),(1101, 'dc01', 'family', 35.00),(1101, 'fb02', 'double', 86.00);
Query OK, 12 rows affected (0.011 sec)
Records: 12 Duplicates: 0 Warnings: 0

MariaDB [hotel_db]> INSERT INTO Booking (HotelNo, GuestNo, DateFrom, DateTo, RoomNo) VALUES('ch01', 10006, '2004-04-21', NULL, 1101),('ch02', 10002, '2004-04-25', '2004-05-06', 801),('dc01', 10003, '2004-05-20', NULL, 1001),('dc01', 10007, '2004-05-13', '2004-05-15', 1001),('fb01', 10001, '2004-04-01', '2004-04-08', 501),('fb01', 10001, '2004-05-01', NULL, 701),('fb01', 10002, '2004-05-04', '2004-05-29', 601),('fb01', 10004, '2004-04-15', '2004-05-15', 601),('fb01', 10005, '2004-05-02', '2004-05-07', 501),('fb02', 10003, '2004-04-05', '2010-04-04', 1001),('fb02', 10005, '2004-05-12', '2030-05-04', 1101);
Query OK, 11 rows affected (0.013 sec)
Records: 11 Duplicates: 0 Warnings: 0

```

b)

i. List the data definition of the tables you have created above.

Query:

SHOW CREATE TABLE Hotel;

SHOW CREATE TABLE Guest;

SHOW CREATE TABLE Room;

SHOW CREATE TABLE Booking;

```

MariaDB [hotel_db]> SHOW CREATE TABLE Hotel;
+-----+
| Table | Create Table
+-----+
| Hotel | CREATE TABLE `hotel` (
 `HotelNo` char(4) NOT NULL,
 `HotelName` varchar(50) NOT NULL,
 `City` varchar(50) NOT NULL,
 PRIMARY KEY (`HotelNo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci |
+-----+
1 row in set (0.001 sec)

MariaDB [hotel_db]> SHOW CREATE TABLE Guest;
+-----+
| Table | Create Table
+-----+
| Guest | CREATE TABLE `guest` (
 `GuestNo` int(11) NOT NULL,
 `GuestName` varchar(50) NOT NULL,
 `GuestAddress` varchar(100) NOT NULL,
 PRIMARY KEY (`GuestNo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci |
+-----+
1 row in set (0.001 sec)

```

```
MariaDB [hotel_db]> SHOW CREATE TABLE Room;
```

```
+-----+
| Table | Create Table
+-----+
| Room | CREATE TABLE `room` (
 `RoomNo` int(11) NOT NULL,
 `HotelNo` char(4) NOT NULL,
 `Type` varchar(20) NOT NULL,
 `Price` decimal(5,2) NOT NULL,
 PRIMARY KEY (`RoomNo`,`HotelNo`),
 KEY `HotelNo` (`HotelNo`),
 CONSTRAINT `room_ibfk_1` FOREIGN KEY (`HotelNo`) REFERENCES `hotel` (`HotelNo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci |
+-----+
1 row in set (0.001 sec)
```

```
MariaDB [hotel_db]> SHOW CREATE TABLE Booking;
```

```
+-----+
| Table | Create Table
+-----+
| Booking | CREATE TABLE `booking` (
 `HotelNo` char(4) NOT NULL,
 `GuestNo` int(11) NOT NULL,
 `DateFrom` date NOT NULL,
 `DateTo` date DEFAULT NULL,
 `RoomNo` int(11) DEFAULT NULL,
 PRIMARY KEY (`HotelNo`,`GuestNo`,`DateFrom`),
 KEY `GuestNo` (`GuestNo`),
 KEY `RoomNo` (`RoomNo`,`HotelNo`),
 CONSTRAINT `booking_ibfk_1` FOREIGN KEY (`HotelNo`) REFERENCES `hotel` (`HotelNo`),
 CONSTRAINT `booking_ibfk_2` FOREIGN KEY (`GuestNo`) REFERENCES `guest` (`GuestNo`),
 CONSTRAINT `booking_ibfk_3` FOREIGN KEY (`RoomNo`,`HotelNo`) REFERENCES `room` (`RoomNo`,`HotelNo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci |
+-----+
1 row in set (0.001 sec)
```

ii. List full details of all hotels.

Query:

```
SELECT * FROM Hotel;
```

```
MariaDB [hotel_db]> SELECT * FROM Hotel;
+-----+
| HotelNo | HotelName | City |
+-----+
ch01	Omni Shoreham	London
ch02	Phoenix Park	London
dc01	Latham	Berlin
fb01	Grosvenor	London
fb02	Watergate	Paris
+-----+
5 rows in set (0.001 sec)
```

iii. List full details of all hotels in London.

Query:



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SELECT \* FROM Hotel WHERE City = 'London';

```
MariaDB [hotel_db]> SELECT * FROM Hotel WHERE City = 'London';
+-----+-----+-----+
| HotelNo | HotelName | City |
+-----+-----+-----+
ch01	Omni Shoreham	London
ch02	Phoenix Park	London
fb01	Grosvenor	London
+-----+-----+-----+
3 rows in set (0.002 sec)
```

iv. List the names and addresses of all guests in London, alphabetically ordered by name.

Query:

SELECT GuestName, GuestAddress

FROM Guest

WHERE GuestAddress LIKE '%London%'

ORDER BY GuestName;

```
MariaDB [hotel_db]> SELECT GuestName, GuestAddress
-> FROM Guest
-> WHERE GuestAddress LIKE '%London%'
-> ORDER BY GuestName;
+-----+-----+
| GuestName | GuestAddress |
+-----+-----+
| John Kay | 56 High St, London |
| Mike Ritchie | 18 Tain St, London |
+-----+-----+
2 rows in set (0.002 sec)
```

v. List all double or family rooms with a price below £40.00 per night, in ascending order of price.

Query:

SELECT \*

FROM Room

WHERE (Type = 'double' OR Type = 'family') AND Price < 40.00

ORDER BY Price;

```

MariaDB [hotel_db]> SELECT *
-> FROM Room
-> WHERE Type IN ('double', 'family') AND Price < 40.00
-> ORDER BY Price ASC;
+-----+-----+-----+-----+
| RoomNo | HotelNo | Type | Price |
+-----+-----+-----+-----+
801	ch02	double	15.00
601	fb01	double	29.00
1001	dc01	double	30.00
1101	dc01	family	35.00
701	fb01	family	39.00
+-----+-----+-----+-----+
5 rows in set (0.001 sec)

```

vi. List the bookings for which no dateto has been specified.

Query:

SELECT \*

FROM Booking

WHERE DateTo IS NULL;

```

MariaDB [hotel_db]> SELECT *
-> FROM Booking
-> WHERE DateTo IS NULL;
+-----+-----+-----+-----+-----+
| HotelNo | GuestNo | DateFrom | DateTo | RoomNo |
+-----+-----+-----+-----+-----+
ch01	10006	2004-04-21	NULL	1101
dc01	10003	2004-05-20	NULL	1001
fb01	10001	2004-05-01	NULL	701
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

```